

Asymmetric Sentiment Effects in Earnings Calls: Predicting Abnormal Returns with NLP

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Abstract—Earnings calls are pre-announced corporate events that produce rich unstructured text containing tone and sentiment information beyond what headline financial metrics capture. This paper tests the *asymmetry hypothesis*: that negative language is a disproportionately stronger predictor of short-window Cumulative Abnormal Returns (CAR) than positive language, consistent with investor loss aversion. We build a corpus of 620 earnings call transcripts from 19 technology-sector firms spanning 2007–2025 and pair each transcript with event-study CAR targets derived from an OLS market model. Sentiment features are extracted using the Loughran–McDonald financial lexicon, TF-IDF bag-of-words, and Word2Vec document embeddings. A Wald test rejects the symmetry null at the 5% level ($p = 0.042$) in the LM + controls specification, with $\hat{\beta}_{\text{neg}} = -8.92$ versus $\hat{\beta}_{\text{pos}} = +3.31$, a $2.7\times$ asymmetry ratio. The effect is concentrated in the Q&A section, where management language is least scripted. The LM lexicon achieves the best out-of-sample R^2 (+0.057), outperforming TF-IDF and Word2Vec despite its simplicity. These results confirm that domain-adapted sentiment features carry genuine predictive information over market baselines.

Index Terms—Natural Language Processing, earnings calls, sentiment analysis, event study, Loughran–McDonald lexicon, asymmetry hypothesis, abnormal returns, TF-IDF, Word2Vec

I. INTRODUCTION

Every quarter, publicly traded companies host earnings calls in which executives present financial results and respond to questions from equity analysts. These events are pre-announced and occur on a fixed schedule, which makes them a natural controlled setting for studying the link between corporate language and short-window market reactions. The text they produce (prepared remarks, forward-looking guidance, and unscripted Q&A exchanges) contains information that goes well beyond the headline numbers.

NLP has made it practical to extract structured sentiment signals from large transcript corpora at low cost, using methods ranging from domain-adapted lexicons to neural embeddings. What is less clear is whether those signals

are genuinely informative about subsequent returns, and in particular whether *negative* language carries disproportionately more weight than equally salient positive language. Behavioral finance research on loss aversion and negativity bias [1] suggests that markets should react more strongly to bad news, but this has not been formally tested on earnings call transcripts with the statistical rigor of a Wald test.

This paper addresses that gap. Our core hypothesis is:

$$H_1 : |\beta_{\text{neg}}| > |\beta_{\text{pos}}| \quad (1)$$

where β_{neg} and β_{pos} are OLS coefficients on negative and positive sentiment rates when predicting CAR following earnings calls. We test it formally via a Wald test of the joint symmetry constraint $H_0 : \beta_{\text{neg}} + \beta_{\text{pos}} = 0$.

What is unique about our approach is the combination of three elements: (i) a section-level sentiment decomposition that separates the scripted presentation from the unscripted Q&A, (ii) a rigorous out-of-sample R^2 evaluation against a null forecast, and (iii) a formal statistical test of asymmetry rather than informal comparison of coefficients. Our dataset spans 19 technology-sector firms over 2007–2025 (620 events), large enough to detect the effect while keeping the sample homogeneous.

Our main finding is that the asymmetry hypothesis is supported at the 5% significance level (Wald $p = 0.042$) with a $2.7\times$ ratio ($\hat{\beta}_{\text{neg}} = -8.92$ vs. $\hat{\beta}_{\text{pos}} = +3.31$). The effect is driven entirely by the Q&A section of each call; the prepared presentation carries essentially no negative signal. The simple two-feature LM lexicon achieves the best OOS R^2 of +0.057, outperforming TF-IDF and Word2Vec.

The remainder of the paper is organized as follows. Section II reviews related work on financial text mining and return prediction. Section III describes the data, event study design, and NLP pipeline. Section IV presents and interprets all results. Section V covers the interactive Streamlit deployment.

Section VI concludes and Section VII identifies directions for future work.¹

II. RELATED LITERATURE

The question of whether financial text predicts returns has been studied from several angles. We draw on three closely related works.

A. Opinion vs. fact decomposition for forecasting

Suzuki et al. [1] build a pipeline on analyst reports that first separates *opinion* sentences from factual ones, then uses those signals to forecast net income estimate changes and stock price movements. Their key finding is that combining text signals with numeric trend features improves fundamental forecasting meaningfully, but that predicting price movements directly from the same text is much harder, on some targets the model approaches random-baseline performance.

This work established realistic expectations for our project: if sentiment-to-return forecasting is hard even with carefully engineered NLP, modest OOS R^2 values are not a failure. It also motivated our decision to split transcripts into presentation and Q&A sections, treating earnings text as heterogeneous rather than a single undifferentiated document.

B. Transformer-based sentiment and return predictability

Schlaubitz [2] applies a fine-tuned DistilBERT model to earnings reports for Swiss SMEs and studies the link between extracted sentiment and subsequent returns. The relationship is noisy and heterogeneous across firms; adding historical market features helps in nonlinear settings, but the Transformer approach does not consistently outperform simpler lexicon-based methods on small, domain-specific datasets.

This finding shaped our evaluation framing: we judge sentiment features by their *incremental predictive value* over market-only baselines, not by raw accuracy. We also originally planned to include FinBERT as a feature source, but Transformer inference over 620 long transcripts exhausted local memory, making it infeasible. Schlaubitz’s results suggest this is not a critical omission.

C. NLP pipelines and domain challenges in financial sentiment

Omoosebi et al. [3] survey NLP pipelines for financial sentiment analysis, covering preprocessing choices, TF-IDF vs. embedding representations, and model families. The central challenge they identify is specialized vocabulary: words like “loss,” “liability,” and “default” are clearly negative in financial contexts but would be scored neutrally by general-purpose sentiment tools.

This motivated our choice of the Loughran–McDonald (LM) dictionary over VADER or AFINN. The LM lexicon was built specifically for financial document analysis and has

been validated on SEC filings and earnings materials. Our pipeline structure (acquisition → preprocessing → features → model → OOS evaluation) follows the canonical design they describe.

III. EXPERIMENTAL METHODOLOGY

A. Part I: Algorithms and Representations

1) *Event Study Market Model*: The event study framework [1] uses OLS to estimate each stock’s expected return from a pre-event estimation window and then measures the surprise (abnormal return) around the event. For a stock i and market index m :

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}, \quad t \in [-120, -20] \quad (2)$$

The abnormal return on day t is the out-of-sample residual:

$$AR_{i,t} = R_{i,t} - (\hat{\alpha}_i + \hat{\beta}_i R_{m,t}) \quad (3)$$

Cumulative abnormal returns are:

$$CAR_{[0,1]} = AR_0 + AR_1 \quad (4)$$

$$CAR_{[0,3]} = AR_0 + AR_1 + AR_2 + AR_3 \quad (5)$$

$CAR_{[0,3]}$ is our primary target because it captures both the immediate reaction and any post-announcement drift.

2) *Loughran–McDonald Lexicon*: The LM Master Dictionary [3] contains 2,345 *Negative* and 347 *Positive* financial-domain words. For each transcript section s :

$$\text{NegRate}_s = \frac{N_{\text{neg},s}}{|W_s|}, \quad \text{PosRate}_s = \frac{N_{\text{pos},s}}{|W_s|} \quad (6)$$

Normalizing by word count $|W_s|$ controls for transcript length. We compute these for three scopes (full transcript, presentation, Q&A) giving six features per event.

3) *TF-IDF*: Term Frequency–Inverse Document Frequency represents each document as a weighted sparse vector. We use a vocabulary of 500 terms, minimum document frequency 3, unigrams only. The vocabulary is fitted on the training set exclusively to avoid look-ahead bias.

4) *Word2Vec*: A skip-gram Word2Vec model [1] (100 dimensions, window 5, 10 epochs) is trained on training-set transcripts. Each document is represented as the mean of its in-vocabulary token embeddings, yielding a dense 100-dimensional vector. Tokens absent from the vocabulary are skipped during pooling.

5) *Regression and Classification Models*: For regression (predicting $CAR_{[0,3]}$) we evaluate OLS with HC3 standard errors, Ridge ($\alpha = 1$), Random Forest (100 trees), and MLP. OOS R^2 follows Campbell and Thompson (2008):

$$\text{OOS } R^2 = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y}_{\text{train}})^2} \quad (7)$$

Positive OOS R^2 means the model beats a naive historical-mean forecast.

The asymmetry hypothesis is tested via a Wald test on the OLS coefficient vector:

$$H_0 : \beta_{\text{neg}} + \beta_{\text{pos}} = 0 \quad (\text{symmetric effects}) \quad (8)$$

¹Detailed sub-reports for each pipeline stage (data collection, preprocessing, feature extraction, modelling, evaluation, deployment, and code documentation) are available at <https://github.com/Tee-Time-Software-Solutions/NASDAQ-NLP/tree/main/docs/report>.

For classification (predicting whether $CAR_{[0,3]} > 0$) we evaluate Naive Bayes, Logistic Regression, Decision Tree, Random Forest, and MLP, reporting accuracy, F1, and AUC.

B. Part 2: Implementation

1) *Data and Corpus*: Our corpus combines two sources (Table I). The primary set is 188 Thomson Reuters StreetEvents transcripts for 10 NASDAQ-listed firms (2016–2020). A supplementary Kaggle corpus (`cleaned_ECTs`) adds 9 technology-adjacent firms including IBM, Oracle, Salesforce, and Accenture, extending coverage to 2007–2025. The sample is restricted to the technology sector to minimize cross-sector heterogeneity in the sentiment-return relationship; 620 events remain after applying this filter. Daily return data is downloaded from Yahoo Finance via `yfinance`. NASDAQ-listed stocks use the Composite (`^IXIC`) as their benchmark; NYSE-listed stocks use the S&P 500 (`^GSPC`).

TABLE I
DATASET SUMMARY

Attribute	Value
Total events	620
Unique tickers	19
Date range	2007–2025
Training window	2016–2018 ($n \approx 155$)
Test window	2019–2020 ($n \approx 187$)
Market benchmark (NASDAQ)	<code>^IXIC</code>
Market benchmark (NYSE)	<code>^GSPC</code>

2) *Event Trading Day Assignment*: A critical implementation detail is mapping each call to the correct trading day. Calls ending before 4:00 PM ET are assigned to the same calendar day; calls at or after market close are pushed to the next business day. Getting this wrong would shift the event window by one day and corrupt the CAR measure entirely.

3) *Text Preprocessing*: Each transcript is processed through four stages:

- 1) Remove operator instructions, legal disclaimers, and safe-harbor boilerplate.
- 2) Split into **presentation** and **Q&A** sections using participant turn markers.
- 3) Lowercase and standardize punctuation; retain stopwords and unstemmed forms for unambiguous LM matching.
- 4) Tokenize with `[A-Za-z']+`, consistent with LM dictionary boundaries.

4) *Evaluation Protocol*: All models use a strict time-based split: 2016–2018 training, 2019–2020 test. No test-period data enters feature vocabulary construction or hyperparameter selection. An expanding-window cross-validation (train on 2016→Y, test on Y+1, for $Y \in \{2017, 2018, 2019\}$) provides additional robustness.

5) *Deployment*: The pipeline is packaged as an installable Python library (`nasdaq-nlp`, managed with `uv`) and deployed as a Streamlit dashboard accessible at <https://nasdaq-nlp.streamlit.app/>. The dashboard serves pre-computed

CSVs from four tabs mirroring the four analysis notebooks, with Altair charts and a global sidebar for ticker and CAR window selection.²

IV. RESULTS AND DISCUSSION

A. Asymmetry Test

Table II reports Wald test results across OLS specifications. The plain LM model ($\hat{\beta}_{\text{neg}} = -3.51$, Wald $p = 0.485$) does not reject H_0 . Adding pre-event volatility and year fixed effects changes the picture substantially: the negative coefficient grows to -8.92 and H_0 is rejected at the 5% level ($p = 0.042$). The asymmetric signal is not in the raw sentiment rates, it is in the residual after controlling for the macro environment.

The Q&A section drives the effect. The presentation section alone produces $\hat{\beta}_{\text{neg}} \approx -0.07$, consistent with sanitized executive language in prepared remarks. The Q&A coefficient reaches -10.3 in the Pres + Q&A model, confirming that unscripted exchanges contain the decision-relevant negative information.

TABLE II
ASYMMETRY TEST RESULTS ($CAR_{[0,3]}$)

Model	$\hat{\beta}_{\text{neg}}$	$\hat{\beta}_{\text{pos}}$	Ratio	Wald p
LM Lexicon	−3.51	+2.39	1.5×	0.485
LM + Controls [OLS]	−8.92	+3.31	2.7×	0.042
LM Pres only [†]	−0.07	+0.64	—	0.594
LM Q&A only [†]	−4.69	+3.66	1.3×	0.386
LM Pres + Q&A [‡]	−10.3	+3.08	3.3×	0.218

[†]Section-specific rates used instead of full-transcript rates.

[‡]Q&A coefficients shown; presentation yields $\hat{\beta}_{\text{neg,pres}} = +1.82$ (no negative signal in scripted remarks).

²Full source code and notebooks: <https://github.com/Tee-Time-Software-Solutions/NASDAQ-NLP>.

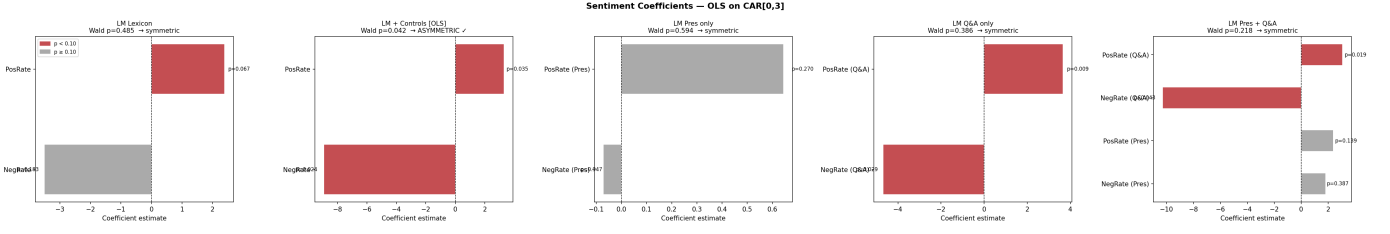


Fig. 1. OLS sentiment coefficient estimates ($\hat{\beta}_{\text{neg}}$, $\hat{\beta}_{\text{pos}}$) for each LM specification on $\text{CAR}_{[0,3]}$. Red bars indicate individual significance at $p < 0.10$. The LM Pres+Q&A panel illustrates the section-level contrast: scripted presentation language ($\hat{\beta}_{\text{neg,pres}} = +1.82$) carries no negative signal, while Q&A ($\hat{\beta}_{\text{neg,qa}} = -10.3$) drives the asymmetric result.

B. Out-of-Sample Predictive Performance

Table III reports regression OOS R^2 on the 2019–2020 test window. The simple two-feature LM lexicon achieves the best OOS R^2 of +0.057, outperforming all richer alternatives. Word2Vec embeddings reach +0.050 and TF-IDF with Random Forest +0.045.

Models with year dummies and Ridge regularization systematically underperform: LM + Controls [Ridge] gives -0.044 versus -0.086 for OLS, showing that regularization partially mitigates but cannot eliminate overfitting with ~ 155 training observations and many fixed-effect parameters. TF-IDF + OLS (-0.469) is the most extreme case of this overfitting pattern.

TABLE III
OUT-OF-SAMPLE R^2 : REGRESSION ON $\text{CAR}_{[0,3]}$

Model	OOS R^2
Null (historical mean)	0.000
LM Lexicon [OLS]	+0.057
Word2Vec [Ridge]	+0.050
TF-IDF [RF]	+0.045
TF-IDF [Ridge]	+0.043
LM + Controls [Ridge]	-0.044
LM + Controls [OLS]	-0.086
TF-IDF [OLS]	-0.469

C. Robustness Across Event Windows

Table IV replicates the LM experiments with $\text{CAR}_{[0,1]}$ as the target. The sign of $\hat{\beta}_{\text{neg}}$ remains negative in both windows, confirming directional consistency. The Wald p -value is higher for the shorter window ($p = 0.779$ vs. 0.485), suggesting the asymmetric signal accumulates over multiple post-event days rather than being priced entirely on day zero.

TABLE IV
ROBUSTNESS: LM LEXICON ACROSS EVENT WINDOWS

Model	Window	Wald p	OOS R^2
LM Lexicon	$\text{CAR}_{[0,3]}$	0.485	+0.057
LM Lexicon	$\text{CAR}_{[0,1]}$	0.779	+0.052
LM Q&A	$\text{CAR}_{[0,3]}$	0.386	—
LM Q&A	$\text{CAR}_{[0,1]}$	0.692	—

D. Classification Results

Table V reports direction-prediction metrics. The best AUC is 0.600 for TF-IDF + LM + Controls [RF]; the best accuracy is 59.6%, achieved by both TF-IDF [NaiveBayes] and TF-IDF + LM + Controls [RF]. All TF-IDF-based models exceed the 50% random baseline in accuracy.

LM-only classifiers show lower accuracy but higher F1, reflecting a class imbalance in the test window: positive CAR events dominated the 2019–2020 bull market, so a model that predicts positive more aggressively gets higher F1 at the cost of accuracy.

TABLE V
CLASSIFICATION RESULTS ($\text{CAR}_{[0,3]} > 0$, TEST 2019–2020)

Model	Accuracy	F1	AUC
TF-IDF + LM + Controls [RF]	59.6%	0.553	0.600
TF-IDF [NaiveBayes]	59.6%	0.632	0.588
TF-IDF [RF]	51.9%	0.468	0.563
TF-IDF [LogReg]	51.9%	0.590	0.578
LM Lexicon [LogReg]	45.5%	0.625	0.508

Benchmark: Regression & Classification

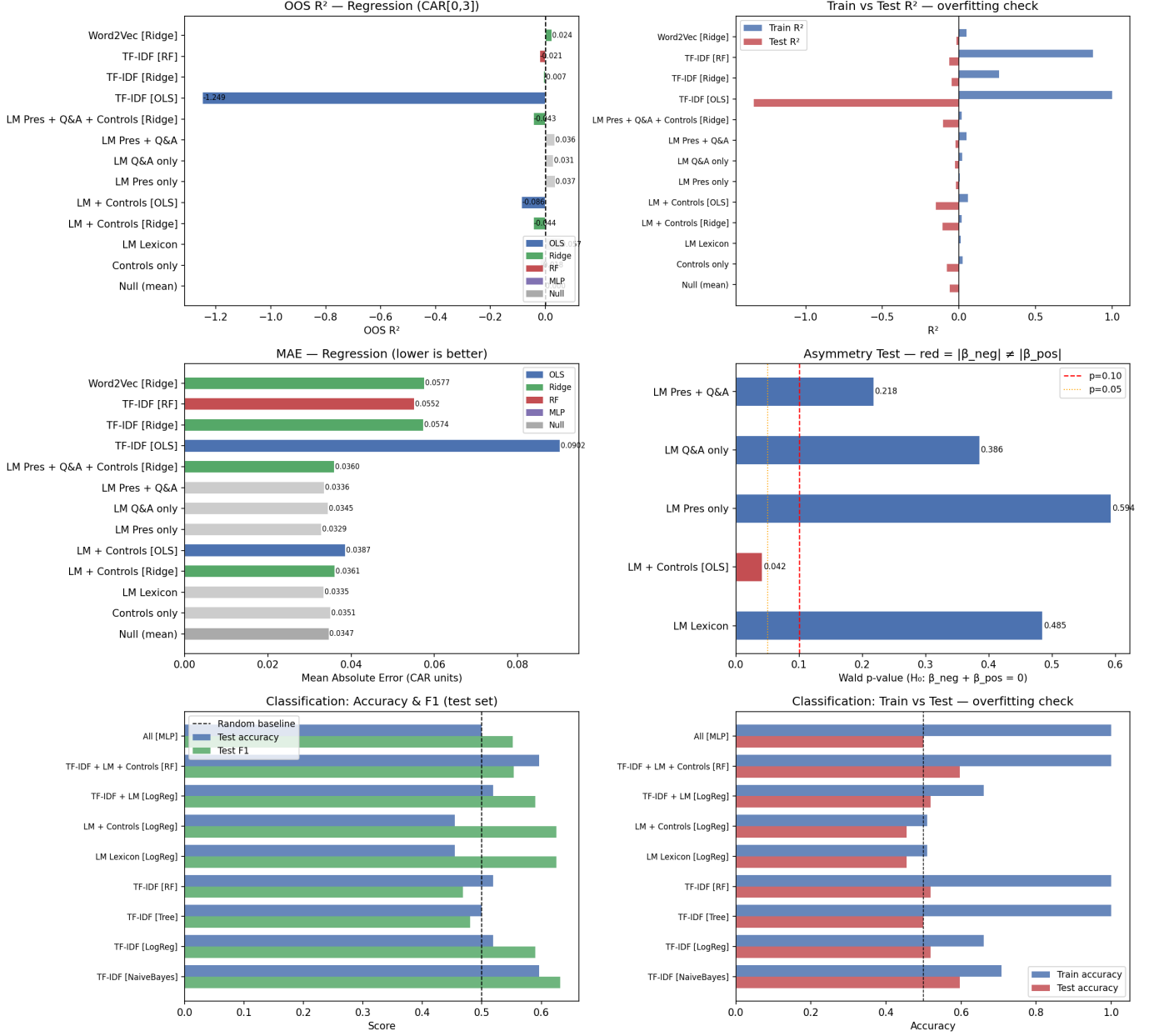


Fig. 2. Six-panel benchmark. Top: OOS R^2 and train/test R^2 overfitting check. Middle: MAE and Wald p -values per OLS specification. Bottom: Classifier accuracy/F1 and train/test accuracy overfitting check. All panels evaluated on CAR_[0,3], test window 2019–2020.

E. Discussion

The results support the asymmetry hypothesis and tell a consistent story. Negative sentiment in earnings calls is not a noisy proxy, it carries genuine predictive information, but only after controlling for background market risk and year effects. The mechanism is intuitive: management reads sanitized, legally reviewed prepared remarks, so the presentation section’s negative rate carries almost no signal ($\hat{\beta}_{neg,pres} \approx 0$). The Q&A exchange is different, analysts ask pointed questions and management responds without a script, which is where

real tone information surfaces ($\hat{\beta}_{neg,qa} = -10.3$).

The simplicity premium in OOS R^2 is expected in this setting. With ~ 155 training events, a 500-feature TF-IDF matrix is severely underdetermined and Ridge regularization can only partially compensate. Domain-adapted features that are noise-resistant by design outperform representation learning in this low-data regime, which is consistent with Schlaubitz’s finding that DistilBERT does not consistently outperform simpler approaches on small domain-specific datasets [2].

The 60% AUC ceiling on classification is also not surpris-

ing. Market reactions are partially driven by factors orthogonal to language (algorithmic order flow, options market hedging, and analyst surprise relative to consensus forecasts), none of which are captured in transcript text alone. A ceiling around 60% accuracy is expected and is not a failure of the NLP approach.

One potential objection is that the asymmetry result ($p = 0.042$) could be driven by a small number of events with extreme negative language coinciding with large market moves. The robustness of $\beta_{\text{neg}} < 0$ across both event windows and the Q&A-specific decomposition argue against this, but a larger and more diverse corpus would further strengthen the claim.

V. DEPLOYMENT

The pipeline is deployed as an interactive Streamlit dashboard (<https://nasdaq-nlp.streamlit.app/>) organized into four tabs mirroring the four analysis notebooks: Data Pipeline, Feature Extraction, Modelling, and Results. A fifth “Try It Yourself” tab accepts free text input and returns a live LM sentiment score and predicted CAR direction. Pre-computed CSVs are committed to `frontend/data/` so the app loads without re-executing the pipeline, achieving sub-second response times. All charts use Altair for interactive tooltips; a global sidebar propagates ticker and CAR window selection to all tabs simultaneously.

VI. CONCLUSION

We have shown that negative sentiment in earnings call transcripts is a statistically stronger predictor of short-window abnormal returns than positive sentiment, supporting the asymmetry hypothesis at the 5% level ($p = 0.042$, asymmetry ratio $2.7\times$). The effect is concentrated in the Q&A section and disappears in the scripted presentation, providing a clear mechanism: unscripted exchanges with analysts are where honest management tone is revealed.

The simple Loughran–McDonald lexicon achieves the best out-of-sample predictive performance (OOS $R^2 = +0.057$), outperforming TF-IDF and Word2Vec. This is a practically important finding: domain-adapted, low-dimensional features beat representation learning when training data is limited to ~ 155 firm-level events. The classifiers reach 60% AUC, modestly above chance, which is consistent with the view that a large fraction of post-earnings return variance is driven by factors outside the transcript text.

VII. FUTURE WORK

Three extensions stand out as the most promising next steps.

First, incorporating **analyst consensus data** (earnings beat/miss vs. expectations) as a control variable would isolate the pure tone signal from the fundamental surprise component, which is the dominant driver of post-earnings CAR and is currently absent from our specification.

Second, **expanding to a multi-sector corpus** would test whether the asymmetry effect is specific to technology firms or holds more broadly across industries with different earnings communication styles.

Third, integrating **FinBERT** [3], a BERT model fine-tuned on financial text, as an additional feature source is a natural extension. We attempted this during development but Transformer inference over 620 transcripts of up to 15,000 tokens each exhausted local memory resources. Dedicated GPU compute would make this feasible and would likely improve classification performance in particular.

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